

An Innovative Secure and Privacy-Preserving Federated Learning-Based Hybrid Deep Learning Model for Intrusion Detection in Internet-Enabled Wireless Sensor Networks

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Abstract—Cyberspace faces numerous security challenges, necessitating advanced research in intrusion detection systems (IDS) to mitigate vulnerabilities. Wireless Sensor Networks (WSNs) connected to the Internet are particularly vulnerable, requiring robust protection mechanisms. Traditional IDS struggle with identifying unknown attacks and maintaining data privacy, especially in WSNs. This study proposes a novel approach integrating Stacked Convolutional Neural Networks (SCNN), Bidirectional Long Short Term Memory (Bi-LSTM), and the African Vulture Optimization Algorithm (AVOA) within a framework of Federated Learning (FL). The integrated model, SCNN-Bi-LSTM-AVOA-FL, aims to enhance intrusion detection efficacy while preserving data privacy. A tailored AVOA optimization method fine-tunes SCNN-Bi-LSTM hyperparameters, leveraging specialized datasets (WSN-DS, CIC-IDS-2017, and WSN-BFSF) for attack detection and categorization. Evaluations compare variants with and without FL techniques (proposed-1 and proposed-2) across metrics such as accuracy, precision, recall, and F1-Score. Results demonstrate significant

improvements in prediction performance, validating the efficacy of the proposed approach in enhancing IDS capabilities for WSNs. This research contributes a comprehensive framework for addressing security challenges in WSNs through advanced machine learning and optimization techniques.

Index Terms—Deep learning, decision making, federated learning, intrusion detection system, wireless sensor network.

I. INTRODUCTION

SENSOR nodes with limited communication and computing capabilities often form part of wireless sensor networks (WSNs), especially in hostile environments. Trusting the information collected from WSN applications, monitoring systems, and early warning systems is important since these applications are critical.

As WSNs grow and fail, sensor nodes are added or removed and replaced due to the dynamic nature of the network. Establishing trust among nodes in sensor networks is even more difficult than in ad-hoc networks. It is, therefore, essential that sensor nodes perform their functions reliably. Sensor nodes deployed in hostile environments may not always behave reliably due to their limitations and limited capabilities [1]. Data breaches, unauthorized access, and disruptions can all be prevented using detection techniques in IoT systems. The model can detect IoT intrusions to improve IoT security since connected devices have tons of data and enough computing power. Wireless sensor networks are important security measures in IoT applications [2]. Detecting malicious activity through network traffic is the task of IDS. An IDS uses the K-means algorithm [3] to detect intrusions because it is an unsupervised learning algorithm [4], [5]. At the same time, support vector machines (SVMs) and k-nearest neighbours are supervised learning algorithms used by IDS. Chaos game optimization algorithms have recently contributed to the advancement of IDS. IDS can be applied to many datasets, but many attributes are redundant and inapplicable. The traditional methods are, therefore, very computationally intensive. This problem can be solved using a variety of optimization algorithms [6]. The Chaos game optimization (CGO) technique is used in the IOT-WSN. CGO can solve

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many optimization problems, including resource allocation, task scheduling, energy management, and network routing [7].

This study aims to model WSN behaviours resulting from IoT devices, including device access control, connection-related activities, firmware upgrades, and remote device access and control. It can be achieved by developing a model that identifies malicious behaviour across the WSN and blocks remote access based on that identification. Detectors designed to operate on Internet-enabled WSN sensor devices must be lightweight and efficient since their computational resources are typically limited. Recent IoT-WSN threats include botnets and DDoS attacks. DDoS attacks targeting large-scale IoT devices will grow in sophistication as the number of connected devices increases. The importance of accurate cybersecurity predictions cannot be overstated. By deploying targeted security measures more effectively, enterprises and individuals minimize unnecessary losses while improving WSN reliability and reducing operational costs. Datasets used to validate accuracy, precision, recall, and F1 scores include WSN-DS, CIC-IDS2017, and WSN-SFBF.

II. LITERATURE REVIEW

Some ensemble methods detect attacks and anomalies, including decision trees (DT), random forests (RF), bagging, boosting, and stacking algorithms [5]. First-time cross-validation, feature selection, and multi-class classification are combined. A stacking ensemble model outperforms others when tested with recent attack datasets, promising robust cybersecurity in smart cities [8]. This study makes use of traditional ML algorithms. It is possible to overlook sophisticated cyber threats in smart city contexts because advanced machine learning models and deep learning approaches have not been investigated. The author [9] examines false data injection attacks (FDIAs) to detect cyberattacks on smart grids with deep learning. It develops an automated, real-time detection and localization system for detecting FDIAs. This architecture allows conventional bad data detectors to be incorporated into the system, providing an effective approach at a lower cost. A multilabel classification strategy and attack co-occurrence dependencies are also available, which can be captured without prior statistical information about the grid.

Each device's behaviour profile is built using deep autoencoders [10]. With statistical traffic features extracted from benign traffic, they train autoencoders. As new traffic observations are applied to a new IoT device, the trained autoencoder has a larger reconstruction error. The reconstruction error would be large if an anomaly occurred, but either approach can approximate a normal traffic pattern. Many detection methods rely on intrusion or anomaly detection principles in an IoT environment with heterogeneous assets and resources.

These threats can be mitigated through a variety of strategies. A network is usually protected by intrusion detection systems (IDS) as an additional layer of security. Due to malware distributors' use of obfuscation techniques, packed binaries, and string modifications, traditional signature-based IDS methods, like blocklists, are currently struggling to keep up with IoT threats [11], [12]. It is difficult for signature-based

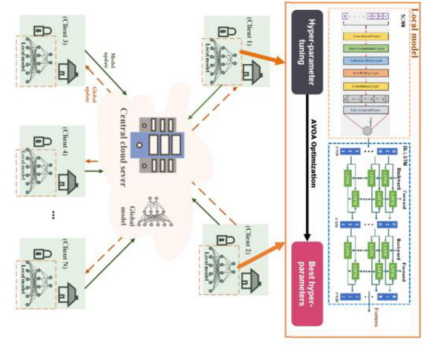


Fig. 1. The systematic layout of the proposed model.

IDS and other methods to keep up with new IoT threats due to rapid changes in botnet control infrastructure and long delays in publishing malware analysis rules. DL and ML technologies are increasingly used in IDS development instead of traditional signature-based methods, enabling more flexibility and generalization [13]. In IoT settings, cloud-based centralized ML training infrastructures face many challenges due to these deployments' massive scale and heterogeneity.

III. PROPOSED MODEL

A. Stacked-Convolutional Neural Network (S-CNN)

Recent developments in 1D-CNNs have modified conventional 2D CNNs [21]. CNNs (convolutional neural feed-forward networks) are among the most well-known deep learning techniques. Figure 1 shows stacked CNN's basic architecture. In most cases, the input is processed using rectified linear units (ReLU) considering layers, neurons, and activation processes. Activation of this type typically adds nonlinearity. When dropout layers are used and the data is normalized (Scaling Data), it is usually possible to prevent overfitting. The computational complexity of 1D-CNNs is considerably lower than that of 2D-CNNs for the same design, network, and hyperparameter combination.

Many 1D-CNN applications (with one or two hidden CNN layers) use networks with 50 or fewer neurons. Especially on portable devices, the miniature format of a 1D-CNN is ideal for real-time, low-cost applications. The 1D-CNNs that were compact and had weak labelling and high signal fluctuations performed better in the specialized study. 1D-CNNs use 1-dimensional arrays instead of matrices as input vectors, as opposed to 2D-CNNs, which use matrices. The following equations are required to work CNN:

$$x_{o,fl}^l = f\left(\sum_{im} x_i^{l-1} * k_{i0,fl}^l + y^l\right) \quad (1)$$

A layer's output is represented by $x_{o,fl}^l$ in Eq. (1), where l is its index. This function $f(\cdot)$ represents a convolution between the input x_i^{l-1} and kernel $k_{i0,fl}^l$, plus bias.

$$x_0^l = f\left[\max\left(\sum_{im} x_i^{l-1}\right) + y^l\right] \quad (2)$$

TABLE I
COMPARISON OF PREVIOUS INTRUSION DETECTION MODEL FOR IOT-WSN

Ref.	Technique	Advantages	Disadvantages
[14]	Layers for transforming and extracting features	False positives are rare due to high accuracy	Data dependency, centralized architecture
[15]	Architecture based on autoencoders and GANs	A low rate of false positives, a high level of accuracy, a high level of true positives	The complexity of the model due to centralized architecture
[16]	DNN	Detection and learning of features in real-time	Input with a high degree of dimensionality, overfitting
[17]	The input is high-dimensional, and overfit	A higher level of detection accuracy while preserving privacy	An efficient and converging communication system
[18]	Efficiency and convergence in communication	Continuous monitoring, improved sensitivity	Scalability and centralization
[19]	CNNs with CLAIRE	Performance and accuracy have been improved	Data requirements for a complex model
[20]	A FL-based intrusion detection system	Identifying behaviour patterns is impossible due to lacking a central repository.	A privacy-preserving, low-overhead communication method

Max pooling in a layer l^{th} is described in Eq. (2). In this example, x_0^l represents the output after the output from layer $(l-1)$ has been max pooled, followed by the bias term y^l and activation function $f(\cdot)$.

$$x_0^l = f(x_i^{l-1} * z_{i0}^l + y^l) \quad (3)$$

Input x_i^{l-1} from the $(l-1)^{th}$ layer is convolved with the kernel z_{i0}^l , adding bias y^l and applying the activation function $f(\cdot)$ to produce output x_0^l .

B. Bidirectional Long Short-Term Memory (Bi-LSTM)

There are different components of an LSTM compared to an RNN. The cell state, which transmits information along the network, is most important in an LSTM [22].

A vector containing values between 0 and 1 is generated by applying the activation function sigmoid, as shown in Figure 1. f_t to the previous hidden state h_{t-1} and the current input x_t . We determine whether we should keep or discard certain information using this layer. Forget gates produce the following output as shown in Eq. (4):

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

The input gate determines cell state information. i_t . Previously hidden states h_{t-1} and current input x_t values make up the input values. As part of the input gate, a sigmoid function selects which information will be used to calculate the candidate cell state. $\tilde{C}_t$, while the candidate's cell state $\tilde{C}_t$ in Eq. (5)& (6) is calculated using a hyperbolic tangent function (tanh). Depending on the importance of the current information, Tanh returns a value between -1 and $+1$. The sigmoid function determines if the current information is important. A candidate cell state and input gate state are calculated as follows:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (6)$$

The result becomes insignificant if f_t is 0. In any other case, f_t is kept if it's 1. This is followed by updating the state of the cell-based in Eq. (7):

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \quad (7)$$

Output gates o_t determine final cell states and hidden states h_t . A sigmoid function is used as input for this gate, while a tanh function is used as input for the current cell state. C_t . A hidden layer carries information based on the sigmoid tanh output and the output, as shown in Eq. (8):

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (8)$$

$$h_t = o_t \times \tanh(C_t) \quad (9)$$

A gate's weights, bias vectors $\tilde{C}_t$, and cell memory is calculated using Eq. (9), where they are W_f , W_i , W_c and W_o are weights; b_f , b_i , b_c , respectively.

There are two bidirectional forms of neurons in Bi-directional RNNs (Bi-RNNs): one for forward states and one for backward states. As a result, data can be input from the past and the future in two time directions. Combining Bi-RNN and LSTM structures, the Bi-LSTM has developed successful applications. A detailed explanation of the Bi-LSTM can be found in [23]. Assume N samples are represented by $S = \{(y^i, x^i)\}_{i=1}^N$. Four features are included in the sample: a three-dimensional path and a timer. A binary hit-miss value is assigned to y_i in the hit-miss classification task. An evaluation of x^{i+l} is y_i for the generation task. Likewise, a direct sequence and an inverse sequence can be concatenated with a Bi-LSTM layer, as shown in the following notation in Eq. (10), (11) & (12):

$$\vec{h}_t = LSTM(x_t, \vec{h}_{t-1}) \quad (10)$$

$$\overleftarrow{h}_t = LSTM(x_t, \overleftarrow{h}_{t-1}) \quad (11)$$

$$y_t = g(W_{\rightarrow hy}, \overleftarrow{h}_t + W_{\leftarrow hy}, \overleftarrow{h}_t + b_y) \quad (12)$$

An LSTM uses $LSTM(\cdot)$ to represent all functions and $g(\cdot)$ to represent activation function, W to represent weight, and b to represent bias.

C. Integrated SCNN-Bi-LSTM

Data with large scales and high fluctuations can be effectively extracted with Stacked Convolutional Neural Networks (SCNN). A SCNN's hierarchical representation allows it to identify complex patterns within data. As the SCNN layers develop, it becomes more capable of detecting intricate patterns of intrusions. In addition to the Bi-LSTM component, the SCNN can easily extract spatial features with the help of the Bidirectional Long Short-Term Memory (Bi-LSTM) component. A comprehensive context provides bidirectional processing with the ability to recognize every temporal pattern within the data. For example, port scanning attacks, which occur over time, are especially beneficial for detecting progressive intrusions. SCNN and Bi-LSTM work synergistically to develop a powerful intrusion detection model. This model detects temporal and spatial intrusion patterns as an Intrusion Detection System (IDS). SCNN-Bi-LSTM's comprehensive feature analysis and efficiency make it a popular choice in IoT networks with limited computational resources. The SCNN-Bi-LSTM architecture is shown in Figure 1.

D. Hyperparameter Tuning Using AVOA

We apply the AVOA technique to the DBiGRU model for optimal hyperparameter tuning, thus improving fault detection accuracy. Some hyperparameters are to be considered here, such as the learning rate, the batch size, and the number of epochs. African vultures are motivated by their foraging and navigational behaviours [24]. The population $n = [1, 2, \dots, N]$ of vultures is represented by N vultures in the environment. After calculating each location position's fitness function, proceed to the next step. The probabilities of selecting the 1st or 2nd group are indicated by p_n

$$P_n = \frac{F_n}{\sum_{n=1}^N F_n} \quad (13)$$

As shown in Eq. (13), in the n^{th} location denotes the fitness function. In each iteration, Eq. (14) can be used to construct 1st or 2nd groups:

$$R(it) = \begin{cases} \text{first group} & P_n = L_1 \\ \text{second group} & P_n = L_2 \end{cases} \quad (14)$$

In contrast, hungry vultures cannot travel long distances for food, while satiated vultures can. Ranges of the data L_1 and L_2 are $0 \leq L_1, L_2 \leq 1$ and $L_1 + L_2 = 1$. The movement from exploration to exploitation can be formulated using Eq. (15) and (16) by determining the rate at which one is hungry or satiated:

$$A = (2 \times \text{rand}_1 + 1) \times x \times \left(1 - \frac{it}{IT_{\max}}\right) + y \quad (15)$$

$$y = h \times \left(\sin^z \left(\frac{\pi}{2} \times \frac{it}{IT_{\max}} \right) + \cos \left(\frac{\pi}{2} \times \frac{it}{IT_{\max}} \right) - 1 \right) \quad (16)$$

x , h , and rand represent random integers that vary from -1 to 1 , -2 to 2 , and 0 to 1 , respectively; This represents the energy of the most energetic vulture, while z represents the possibility of entering exploration; it is the current iteration and $it_{\max}$ is the maximum iteration. In the exploration phase,

the African vulture determines which strategy to use according to the parameter. p_1 , $0 \leq p_1 \leq 1$ in Eq. (17-19).

$$P(it+1) = \begin{cases} (8), & p_1 \geq \text{rand}_2 \\ (13), & p_1 < \text{rand}_2 \end{cases} \quad (17)$$

$$P(it+1) = R(it) - D(it) \times A \quad (18)$$

$$D(it) = |q \times R(it) - P(it)| \quad (19)$$

A vulture's location vector is indicated by $P(it+l)$ in the next iteration. As A is attained and $R(it)$ means 1, $q = 2 \times \text{rand}_3$ is attained. rand_3 , A' , it ranges from 0 to 1 at random in eq. (20)

$$P(it+l) = R(it) - A + \text{rand}_4 \times ((u_b - 1) \times \text{rand}_5 + 1_b) \quad (20)$$

rand_4 and rand_5 show the random integers within $[0, 1]$ in Eq. (21), where u_b and 1_b indicate upper and lower boundaries. There are two stages to the exploitation phase. The variables p_2 and p_3 play a crucial role in the selection of any strategy. A range of values exists between $[0,1]$ and $[0,1]$. In the exploitation phase, if $|F|$ lies between $[1, 0.5]$, siege fight and rotating flight are the two initial strategies:

$$P(it+1) = \begin{cases} (15), & p_2 \geq \text{rand}_6 \\ (16), & p_2 < \text{rand}_6 \end{cases} \quad (21)$$

The random integer rand_6 is defined as an integer within the range $[0, 1]$ in Eq. (22). As shown below, Eq. (23) can be solved as follows:

$$P(it+l) = D(it) - A + \text{rand}_7 - d(it) \quad (22)$$

$$D(it) = R(it) - P(it) \quad (23)$$

As a next step, we can model the rotational flight of vultures as shown in Eq. (24), (25) & (26):

$$P(it+l) = R(it) - (M_1 + M_2) \quad (24)$$

$$M_1 = R(it) \times \left(\frac{\text{rand}_8 \times P(it)}{2\pi} \right) \times \cos(P(it)) \quad (25)$$

$$M_2 = R(it) \times \left(\frac{\text{rand}_9 \times P(it)}{2\pi} \right) \times \sin(P(it)) \quad (26)$$

A random integer within the range of $[0, 1]$ is denoted by rand_8 and rand_9 in Eq. (27)–(30). If $|F| > 0.5$ occurs, vultures use two distinct approaches for finding food during the exploitation stage: aggressive fights and accumulations of prey. Any strategy can be selected based on the following conditions:

$$P(it+1) = \begin{cases} (12), & p_3 \geq \text{rand}_7 \\ (24), & p_3 < \text{rand}_7 \end{cases} \quad (27)$$

where

$$P(it+1) = \frac{B_1 + B_2}{2} \quad (28)$$

$$B_1 = \text{Best}_{\text{vulture1}}(it) - \frac{\text{Best}_{\text{vulture1}}(it) \times P(it)}{\text{Best}_{\text{vulture1}}(it) \times P(it)^2} \times A \quad (29)$$

$$B_2 = \text{Best}_{\text{vulture2}}(it) - \frac{\text{Best}_{\text{vulture2}}(it) \times P(it)}{\text{Best}_{\text{vulture2}}(it) \times P(it)^2} \times A \quad (30)$$

In the existing iteration, $\text{Best}_{\text{vulture1}}$ and $\text{Best}_{\text{vulture2}}(it)$ represents the optimal vulture for groups 1st and 2nd, respectively.

Here are some examples of aggressive competition among vultures as shown in Eq. (31), (32):

$$P(it+1) = R(it) - |d(it)| \times A \times Levy(d) \quad (31)$$

An optimization problem has dimension d_1 :

$$Levy(x) = 0.01 \times \frac{u \times \sigma}{|V|^{\frac{1}{\beta}}}, \sigma = \left(\frac{Y(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\gamma(1+2\beta) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}} \right)^{\frac{1}{\beta}} \quad (32)$$

The AVOA method relies heavily on fitness selection. Solution encoding is used to evaluate a candidate's goodness (aptitude). Fitness functions are designed based on the accuracy value in Eq. (33) & (34).

$$Fitness = \max(P) \quad (33)$$

$$P = \frac{TP}{TP + FP} \quad (34)$$

A true positive is indicated by TP , and a false positive is indicated by FP .

E. Federated Learning (FL)

In Federated Learning (FL), the preservation of unbiased data representation and privacy is ensured through advanced ML architectures [25]. Each iteration of the model uses local data from the client site. Data centres, smartphones, IoT devices, wearables, and different organizations' data warehouses may all serve as clients. The centralized server aggregates model updates from the local sites without utilizing unprocessed data. Along with smart homes and smart cities, FL is being adopted in the industry and cyber security. As a result of aligning the parameters of the client model with those of the global training model, an aggregated model is developed. In contrast to the server's shared model, clients are trained collaboratively [26]. In comparison to classical ML models, FL is more secure, has compressed latency, and consumes less power [27]. ML models are also customized for different client nodes, so users of different nodes can interact more effectively with the system [28].

The mathematical representation carries X numbers of active clients carrying X numbers of active clients carrying X numbers. Every client's data distribution X is represented by D_X , while the number of specimens available to each client is represented by N_X . The complete specimen size is $N = \sum_{x=1}^X \binom{N}{x} N_x$ in Eq. (35). A FL problem can be used to solve experimental risk minimization problems:

$$\text{Min}_{l \in R^d} F(l) := \sum_{x=1}^X \frac{N_x}{N} F_x(l) \text{ where } F_x(l) := \frac{1}{N_x} \sum_{a_i \in D_X} F_i(l) \quad (35)$$

where, $(a_i \ b_i)$ is an input-output data pair that represents the loss function f_i . $a_i \in R^d$ and $b_i \in R$ or b_i and or here are between -1 and 1 . FL mathematical models are subject to statistics, communications, and security challenges when implemented in various application domains. FL has several advantages in various applications due to its technical features, including safeguarding user privacy, improving the model's efficiency and scalability, and reducing data transfer costs.

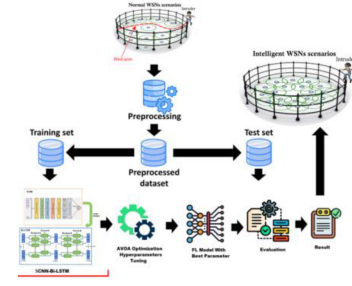


Fig. 2. The experimental layout of the proposed model.

F. Integrated FL-SCNN-Bi-LSTM

Our study aims to improve network intrusion categorization using information from numerous nodes with the help of an FL framework. The basic structure of the proposed model can be seen in Figure 1, where there is one server and one client computer with limited computing capabilities. Each client maintains its local model, while the server shares a global neural network with all clients. During each training round, clients receive the most recent model parameters. Using FL techniques, the proposed model is illustrated in Figure 1. Over multiple rounds, different subsets of clients are iterated over the global model to achieve the ideal state.

IV. RESULT ANALYSIS AND DISCUSSION

This section describes experimental results for FL-SCNN-Bi-LSTM integrated with AVOA optimization-based intrusion detection framework, as shown in Figure 2. Based on Ubuntu, the simulations use a 2.50GHz CPU and Intel Core i7 with 16GB RAM. Three publicly available databases are used for evaluations, including WSN-DS, CIC-IDS2017, and WSN-BFSF Dataset.

Performance metrics are evaluated to validate the proposed model's performance. On the WSN-DS and CIC-IDS2017 datasets, proposed-1 and proposed-2 are compared to existing Support Vector Machines (SVM), K-Nearest Neighbor (KNN), Random Forests (RFs), Neural Networks (NNs), Light Gradient-Boosting Machines (LightGBMs), and LSTMs, BiLSTMs, CNN-LSTM, CNN-GRUs, and GRU-CNNs on WSN-BFSF datasets. As a result of their robust performance and high prediction accuracy, the proposed-1 model and the proposed-2 model were both very successful. The proposed dataset is described in detail below.

WSN-DS Dataset: This work takes it into account. In WSN, WSN-DS is used explicitly to sleuth out DoS incidents. In total, 374,661 archives are included, with four different kinds of DoS attacks indicated: Blackhole, greyhole, Flooding, Scheduling, and normal behaviour (no-attack) [29], as shown in Table II.

CIC-IDS2017 Dataset: Information security data scientists use the Canadian Institute for Cybersecurity Intrusion Detection Evaluation dataset (CIC-IDS 2017) heavily for intrusion detection in the information security field. A CIC-IDS team initiated normal traffic between victims to generate "benign" network traffic. With publically available tools, they launched 14 well-known attack types. As a final step, they

TABLE II
WSN-DS DATASET CLASS LABEL DESCRIPTION

Attack Type	Number of Records
Black hole Attack	14,596
Grey hole attack	10,049
Flooding Attack	6638
Scheduling Attack	3312
Normal	340,066
Total	374,661

TABLE III
THE NUMBER OF CLASSES AND DATA DISTRIBUTION

Class	Row Distribution	Percentage
Normal	262,851	84.22%
Flooding	29,844	9.56%
Blackhole	11,766	3.77%
Selective Forwarding	7,645	2.45%
Total	312,106	100%

saved all PCAP data on traffic flow. CICFlowMeter analysis results are included, as well as labelled flows based on time stamps, source and destination IP addresses, source and destination ports, protocols, and attacks (CSV file). The data collection period spanned five days, starting Monday, July 3, 2017 at 9 a.m. and ending Friday, July 7, 2017 at 5 p.m. The traffic on Monday is benign and is typical for the day. A range of attacks were implemented, including brute force FTP, SSH, DDoS, Web Attack, infiltration, botnets and DDoS.

WSN-BFSF Dataset: This dataset has three types of attacks: blackholes, flooding, and selective forwarding. This dataset also has data from the Selective Forwarding attack, unlike those in the WSN-DS dataset. Moreover, the study will contribute to developers' analysis of Selective Forwarding attack data. After preprocessing, 312,106 rows of data were obtained from the dataset containing 16 features. An instance is represented by a row with a different value for each feature. Several categories of attack traffic are shown in the dataset, including Normal, Flooding, Blackhole, and Selective Forwarding. Table III shows the data distribution by traffic class in some instances for each class.

A. Performance Metrics

There are four classifications of results used for result evaluation: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). There are two types of classifications for data packets: secured and malicious. The model detected an attack successfully, which means malicious packets were detected, and FP means secured packets were detected. Still, normal packets were classified as abnormal. The TN indicates that secured packets were detected as secured packets, while the FN indicates that some malicious packets were classified as secured. The accuracy, precision, recall, and F1-score are used to evaluate the proposed model [5]. The accuracy, recall, precision and F1-Score formulas are mentioned in the Eq. (36)-(39).

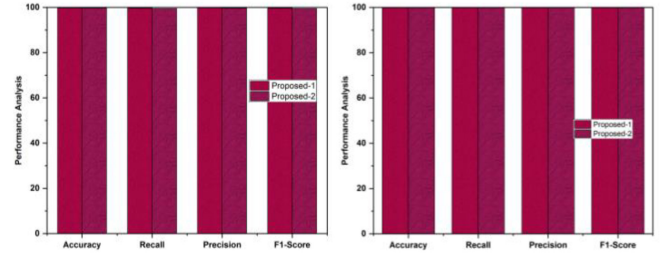


Fig. 3. Performance analysis of the proposed-1 and proposed-2 on WSN-DS (left) and CIC-IDS-2017 (right) datasets.

$$Accuracy = \frac{TP + TN}{(TP + TN + FP + FN)} \quad (36)$$

$$Recall = \frac{TP}{TP + FN} \quad (37)$$

$$Precision = \frac{TP}{(TP + FP)} \quad (38)$$

$$F = \frac{2 * (Precision * Recall)}{(Precision + Recall)} \quad (39)$$

B. Experimental Discussion

In Figure 2, we present the comparative accuracy, recall, precision, and F1 scores for the proposed-1 and proposed-2 versions of the WSN-DS. The proposed-1 model has the performance of accuracy, recall, precision, and F1-Score is 99.65%, 99.98%, 99.58% and 99.77%, and the proposed-2 model shows 99.5%, 99.94%, 99.48% and 98.26% respectively. Compared to the proposed-2 technique, the proposed-1 technique achieves better performance for each network attack. The proposed-1 shows 99.65% higher detection accuracy than proposed-2 (99.50%). The proposed-1 and proposed-2 models consistently perform well across diverse network attack scenarios, proving their effectiveness and reliability as intrusion detection models in the WSN-IoT environment. WSN-IOT common attacks are contained in the CIC-IDS2017 dataset. The proposed-1 and proposed-2 show the higher detection accuracy on the CIC-IDS2017 dataset, as shown in Figure 3. The proposed-1 has 99.93% accuracy, 99.92% recall, 99.93% precision, and 99.93% F1-Score. The integrated FL-SCNN-Bi-LSTM with optimized hyperparameter by AVOA optimization (proposed-1) achieved a higher detection rate than the SCNN-Bi-LSTM with AVOA optimization (proposed-2).

Figure 4 depicts the proposed model's accuracy, recall, precision and F1-Score analysis on the WSN-BFSF dataset. Observing Figure 4, the proposed-1 model attains maximal accuracy over the proposed-2 model. The proposed-1 model attains 99.93%, 99.92%, 99.93% and 99.93% average performance value and the proposed-2 99.90%, 99.91%, 99.90% and 99.9%, respectively. WSN-BFSF's network attacks show improved performance when using the proposed-1 model.

As shown in Figure 5, existing deep learning models such as SVM, KNN, RF, NN, and LightGBM perform equally well as the proposed model. Both models are more accurate, precise, and recallable, with a higher F1-Score indicator than

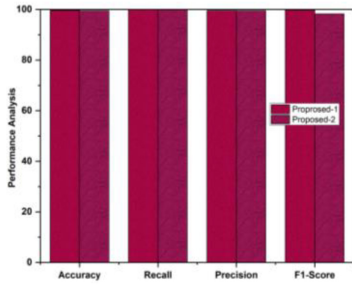


Fig. 4. Performance analysis of the proposed-1 and proposed-2 on the WSN-BFSF dataset.

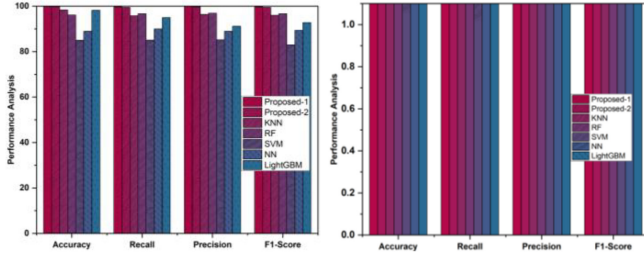


Fig. 5. Comparative state-of-art analysis of WSN-DS (left) and CIC-IDS-2017 (right) dataset.

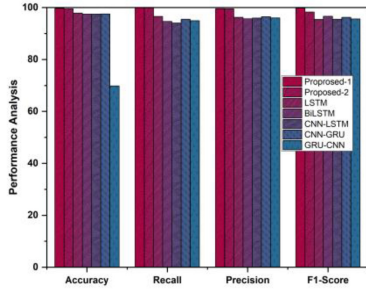


Fig. 6. Comparative state-of-art analysis of the WSN-BFSF dataset.

existing models. Using an integrated FL model with SCNN-Bi-LSTM-AVOA improves accuracy, reduces false alarm rates, and reduces traffic data dimension. Compared to existing feature selection and classification algorithms, the proposed model solves problems such as poor real-time detection, poor detection performance and high model complexity in wireless sensor networks. WSN-DS and CIC-IDS-2017 datasets can be handled with the proposed 1 model due to their high real-time performance and good detection performance.

Figure 6 provides a detailed performance analysis of the proposed-1 model and a comparison with various competing attack detection models based on the WSN-BFSF dataset. A bar plot illustrates the average values of several performance metrics for each model, which are evaluated based on several metrics. This dataset has a mixture of various attack types and normal network traffic, making it an appropriate benchmark for testing different detection methods.

In the proposed-1 model, accuracy is 99.93%, and in the proposed-2 model, accuracy is 99.90%. Based on the proposed-1 model, most instances can be correctly classified. Even a small misclassification rate can result in significant

security breaches in IOT-enabled WSNs with high accuracy levels. A 99.94% precision is achieved by the proposed-1 model, demonstrating its proficiency in identifying normal network traffic. As a result, false alarms are minimized, and unnecessary intervention or resource usage is reduced in IOT-WSNs. A 99.92% recall value is also demonstrated by proposed 1. By detecting and identifying true positives, the model increases network security by detecting malicious activities and attacks. A harmonic mean of Precision and Recall of 99.93% confirms the model's balanced performance. According to the F1-Score, proposed-1 can accurately identify both positives and negatives with a high degree of accuracy.

V. CONCLUSION

The FL-SCNN-Bi-LSTM-AVOV intrusion detection model developed for IoT-based Wireless Sensor Networks (WSNs) represents a significant advancement in securing Internet-enabled environments. Our proposed model effectively detects and categorizes intrusions, ensuring robust security and reliability for WSNs. It outperforms existing methods in terms of accuracy, recall, precision, and F1-Score, demonstrating its superior capability to adapt to evolving attack patterns across diverse WSN environments. As cyber threats continue to evolve, future research will focus on expanding the model's applicability to various IoT-WSN scenarios and integrating with existing security frameworks. This will involve exploring new datasets, evolving attack scenarios, and enhancing scalability. Furthermore, integrating blockchain technology holds promise for further bolstering the scalability and security of our proposed solution, particularly in complex IoT ecosystems like smart grids. The FL-SCNN-Bi-LSTM-AVOV model represents a pivotal step towards enhancing the security posture of IoT-based WSNs. By leveraging advanced machine learning techniques and optimization strategies, this research contributes to the ongoing efforts to safeguard critical infrastructure and ensure the integrity of data transmission in interconnected environments.

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